

Problem Set 2: Classical Mechanics 1 (Aug 2026)

2.4 ★★ The origin of the quadratic drag force on any projectile in a fluid is the inertia of the fluid that the projectile sweeps up. (a) Assuming the projectile has a cross-sectional area A (normal to its velocity) and speed v , and that the density of the fluid is ϱ , show that the rate at which the projectile encounters fluid (mass/time) is ϱAv . (b) Making the simplifying assumption that all of this fluid is accelerated to the speed v of the projectile, show that the net drag force on the projectile is ϱAv^2 . It is certainly not true that all the fluid that the projectile encounters is accelerated to the full speed v , but one might guess that the actual force would have the form

$$f_{\text{quad}} = \kappa \varrho A v^2 \quad (2.84)$$

where κ is a number less than 1, which would depend on the shape of the projectile, with κ small for a streamlined body, and larger for a body with a flat front end. This proves to be true, and for a sphere the factor κ is found to be $\kappa = 1/4$. (c) Show that (2.84) reproduces the form (2.3) for f_{quad} , with c given by (2.4) as $c = \gamma D^2$. Given that the density of air at STP is $\varrho = 1.29 \text{ kg/m}^3$ and that $\kappa = 1/4$ for a sphere, verify the value of γ given in (2.6).

2.11 ★★ Consider an object that is thrown vertically up with initial speed v_0 in a linear medium. (a) Measuring y upward from the point of release, write expressions for the object's velocity $v_y(t)$ and position $y(t)$. (b) Find the time for the object to reach its highest point and its position y_{max} at that point. (c) Show that as the drag coefficient approaches zero, your last answer reduces to the well-known result $y_{\text{max}} = \frac{1}{2} v_0^2 / g$ for an object in the vacuum. [Hint: If the drag force is very small, the terminal speed is very big, so v_0/v_{ter} is very small. Use the Taylor series for the log function to approximate $\ln(1 + \delta)$ by $\delta - \frac{1}{2}\delta^2$. (For a little more on Taylor series see Problem 2.18.)]

2.12 ★★ Problem 2.7 is about a class of one-dimensional problems that can always be reduced to doing an integral. Here is another. Show that if the net force on a one-dimensional particle depends only on position, $F = F(x)$, then Newton's second law can be solved to find v as a function of x given by

$$v^2 = v_0^2 + \frac{2}{m} \int_{x_0}^x F(x') dx'. \quad (2.85)$$

[Hint: Use the chain rule to prove the following handy relation, which we could call the “ $v dv/dx$ rule”: If you regard v as a function of x , then

$$\dot{v} = v \frac{dv}{dx} = \frac{1}{2} \frac{dv^2}{dx}. \quad (2.86)$$

Use this to rewrite Newton's second law in the separated form $m d(v^2) = 2F(x) dx$ and then integrate from x_0 to x .] Comment on your result for the case that $F(x)$ is actually a constant. (You may recognise your solution as a statement about kinetic energy and work, both of which we shall be discussing in Chapter 4.)

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2.20 ** [Computer] Use suitable graph-plotting software to plot graphs of the trajectory (2.36) of a projectile thrown at 45° above the horizontal and subject to linear air resistance for four different values of the drag coefficient, ranging from a significant amount of drag down to no drag at all. Put all four trajectories on the same plot. [Hint: In the absence of any given numbers, you may as well choose convenient values. For example, why not take $v_{x0} = v_{y0} = 1$ and $g = 1$. (This amounts to choosing your units of length and time so that these parameters have the value 1.) With these choices, the strength of the drag is given by the one parameter $v_{\text{ter}} = \tau$, and you might choose to plot the trajectories for $v_{\text{ter}} = 0.3, 1, 3$, and ∞ (that is, no drag at all), and for times from $t = 0$ to 3. For the case that $v_{\text{ter}} = \infty$, you'll probably want to write out the trajectory separately.]

2.21 *** A gun can fire shells in any direction with the same speed v_0 . Ignoring air resistance and using cylindrical polar coordinates with the gun at the origin and z measured vertically up, show that

the gun can hit any object inside the surface

$$z = \frac{v_0^2}{2g} - \frac{g}{2v_0^2} \rho^2.$$

Describe this surface and comment on its dimensions.

2.22 *** [Computer] The equation (2.39) for the range of a projectile in a linear medium cannot be solved analytically in terms of elementary functions. If you put in numbers for the several parameters, then it *can* be solved numerically using any of several software packages such as Mathematica, Maple, and MatLab. To practice this, do the following: Consider a projectile launched at angle θ above the horizontal ground with initial speed v_0 in a linear medium. Choose units such that $v_0 = 1$ and $g = 1$. Suppose also that the terminal speed $v_{\text{ter}} = 1$. (With $v_0 = v_{\text{ter}}$, air resistance should be fairly important.) We know that in a vacuum, the maximum range occurs at $\theta = \pi/4 \approx 0.75$. **(a)** What is the maximum range in a vacuum? **(b)** Now solve (2.39) for the range in the given medium at the same angle $\theta = 0.75$. **(c)** Once you have your calculation working, repeat it for some selection of values of θ within which the maximum range probably lies. (You could try $\theta = 0.4, 0.5, \dots, 0.8$.) **(d)** Based on these results, choose a smaller interval for θ where you're sure the maximum lies and repeat the process. Repeat it again if necessary until you know the maximum range and the corresponding angle to two significant figures. Compare with the vacuum values.

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2.38 ** A projectile that is subject to quadratic air resistance is thrown vertically *up* with initial speed v_o . **(a)** Write down the equation of motion for the upward motion and solve it to give v as a function of t . **(b)** Show that the time to reach the top of the trajectory is

$$t_{\text{top}} = (v_{\text{ter}}/g) \arctan(v_o/v_{\text{ter}}).$$

(c) For the baseball of Example 2.5 (with $v_{\text{ter}} = 35$ m/s), find t_{top} for the cases that $v_o = 1, 10, 20, 30$, and 40 m/s, and compare with the corresponding values in a vacuum.

2.41 ** A baseball is thrown vertically up with speed v_o and is subject to a quadratic drag with magnitude $f(v) = cv^2$. Write down the equation of motion for the upward journey (measuring y vertically *up*) and show that it can be rewritten as $\dot{v} = -g[1 + (v/v_{\text{ter}})^2]$. Use the “ $v dv/dx$ rule”

(2.86) to write $\dot{v}$ as $v dv/dy$, and then solve the equation of motion by separating variables (put all terms involving v on one side and all terms involving y on the other). Integrate both sides to give y in terms of v , and hence v as a function of y . Show that the baseball’s maximum height is

$$y_{\text{max}} = \frac{v_{\text{ter}}^2}{2g} \ln \left(\frac{v_{\text{ter}}^2 + v_o^2}{v_{\text{ter}}^2} \right). \quad (2.89)$$

If $v_o = 20$ m/s (about 45 mph) and the baseball has the parameters given in Example 2.5 (page 61), what is y_{max} ? Compare with the value in a vacuum.

2.43 *** [Computer] The basketball of Problem 2.31 is thrown from a height of 2 m with initial velocity $\mathbf{v}_o = 15$ m/s at 45° above the horizontal. **(a)** Use appropriate software to solve the equations of motion (2.61) for the ball’s position (x, y) and plot the trajectory. Show the corresponding trajectory in the absence of air resistance. **(b)** Use your plot to find how far the ball travels in the horizontal direction before it hits the floor. Compare with the corresponding range in a vacuum.